Classification Model Evaluation Report

Breast Cancer Diagnosis — AI/ML Task 4

1. Overview

This report evaluates two classification models — Logistic Regression and a Decision Tree Classifier — trained on the scikit-learn Breast Cancer Wisconsin dataset (569 samples, 30 numeric features, binary target where 0 = malignant and 1 = benign). The dataset was split 80/20 into training (455 samples) and test (114 samples) sets. The discussion below covers which evaluation metrics matter most for this problem, how class imbalance affects the results, and which model is recommended as the final choice.

2. Metric Selection

2.1 Why Accuracy Alone Is Not Enough

Both models achieve high accuracy (95.6% for Logistic Regression and 94.7% for the Decision Tree), but accuracy treats every misclassification equally. In a cancer-screening context, the two types of errors are not equally costly: a **false negative** (a malignant tumor predicted as benign) could delay treatment for a real cancer case, while a **false positive** (a benign tumor predicted as malignant) typically leads to additional testing. For this reason, accuracy was supplemented with precision, recall, F1-score, and ROC-AUC.

2.2 Confusion Matrix and Recall on the Malignant Class

The confusion matrix for Logistic Regression on the test set was:

	Predicted: Malignant (0)	Predicted: Benign (1)
Actual: Malignant (0)	39	4
Actual: Benign (1)	1	70

Out of 43 malignant cases, 4 were misclassified as benign (recall for the malignant class = 39/43 ≈ 0.91, matching the classification report). These 4 false negatives are the most clinically important errors in this confusion matrix — this is why, for this task, **recall on the malignant class (and the related metric, the false-negative count) is treated as the primary metric**, with precision, F1, and overall accuracy used as supporting metrics.

2.3 ROC Curve and AUC

The ROC curve for Logistic Regression plots the true positive rate against the false positive rate across all classification thresholds, with an AUC of **0.998**. An AUC this close to 1.0 indicates the model separates the two classes almost perfectly across thresholds, and that the default 0.5 threshold is not the only viable operating point — the threshold could be lowered slightly to trade a little precision for higher recall on the malignant class if that trade-off is clinically preferred.

3. Imbalance Handling

The dataset has a moderate class imbalance: 212 of 569 samples (37.3%) are malignant and 357 (62.7%) are benign. This is not an extreme imbalance, but it is large enough that a model could achieve a deceptively high accuracy simply by being biased toward predicting the majority (benign) class.

- **Train/test split:** the current split (80/20, random_state=42) was not stratified. By chance it preserved a similar class ratio in the test set (43 malignant / 71 benign ≈ 37.7% / 62.3%), but using stratify=y in train_test_split would guarantee this in general and is recommended as a best practice.
- **Class-weighted models:** setting class_weight='balanced' in LogisticRegression and DecisionTreeClassifier would up-weight the minority (malignant) class during training, directly targeting the false-negative problem identified in Section 2.2.
- **Resampling:** techniques such as SMOTE (oversampling the malignant class) or random undersampling of the benign class are alternatives, though with 212 malignant samples already available, class weighting is usually sufficient and avoids the risk of overfitting to synthetic samples.
- **Metric choice as imbalance handling:** reporting precision, recall, and F1 per class (as the classification report does) rather than relying on overall accuracy is itself a form of imbalance handling, since it surfaces how the minority class is performing independent of the majority class's larger share of the data.

In the current notebook, no explicit imbalance-handling technique (stratification, class weights, or resampling) was applied. The 4 missed malignant cases for Logistic Regression suggest that adding class_weight='balanced' and re-evaluating recall on the malignant class would be a worthwhile next step.

4. Model Comparison and Final Model Decision

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.9561	0.9459	0.9859	0.9655	0.9977
Decision Tree	0.9474	0.9577	0.9577	0.9577	N/A*

**ROC-AUC was only computed for Logistic Regression in the current notebook; computing predict_proba and roc_auc_score for the Decision Tree would allow a direct comparison.*

4.1 Interpreting the Comparison

Logistic Regression edges out the Decision Tree on accuracy (95.6% vs 94.7%), recall (0.986 vs 0.958) and F1-score (0.966 vs 0.958), and additionally provides probability estimates with a near-perfect ROC-AUC of 0.998. The Decision Tree has a marginally higher precision (0.958 vs 0.946), but its lower recall and lack of calibrated probabilities make it harder to tune toward the low-false-negative behavior that matters most for this task.

4.2 Final Model Recommendation

Logistic Regression is recommended as the final model for the following reasons:

- It achieves the best overall accuracy, recall, and F1-score of the two models evaluated.
- Its near-perfect ROC-AUC (0.998) shows excellent separation between malignant and benign cases across all thresholds, and gives a tunable decision threshold if the cost of false negatives needs to be reduced further.
- As a simpler, linear model, it is less prone to the kind of overfitting that unconstrained Decision Trees are susceptible to, which supports more stable performance on unseen data.

4.3 Limitations and Next Steps

Before finalizing this choice, it is recommended to: (1) add a confusion matrix and ROC/AUC for the Decision Tree so both models are compared on identical metrics; (2) re-run both models with class_weight='balanced' and/or a stratified split to address the class imbalance noted in Section 3; and (3) use 5-fold cross-validation rather than a single train/test split to confirm that the ~1 percentage-point gap between the two models is consistent and not an artifact of this particular split.